

B.S. 4 Years Program / Eighth Semester-2019

Introduction to Economics Course Code: IT-408 Part-1 Time: 2 Hrs. 45Min. Marks: 50

Attempt this (subjective) on the separate answer sheet provided.

Short Answer Questions

1. Define Economics. Write down the difference between Micro and Macro Economics.

Economics is the social science that studies the production, distribution, and consumption of goods and services. It analyzes how individuals, businesses, and governments make choices about allocating scarce resources.

- **Microeconomics** focuses on the behavior of individual economic agents, such as consumers, producers, and firms. It examines how markets work, how prices are determined, and how resources are allocated.
- **Macroeconomics** deals with the economy as a whole, analyzing aggregate variables like GDP, inflation, unemployment, and interest rates. It studies the factors that influence the overall economic performance of a nation.

2. State the Law of Diminishing Marginal Utility.

The **Law of Diminishing Marginal Utility** states that as a consumer consumes more units of a good or service, the additional satisfaction (or utility) they derive from each additional unit decreases. In other words, the more you consume of something, the less valuable each additional unit becomes.

3. What is Consumer Equilibrium?

Consumer Equilibrium occurs when a consumer allocates their income in a way that maximizes their total utility, given their budget constraint. This means that they are purchasing the optimal combination of goods and services to satisfy their wants and needs within their budget.

4. Write down the concept of Perfect Competition.

Perfect Competition is a market structure characterized by:

- Many small buyers and sellers
- Homogeneous products
- Perfect information
- No barriers to entry or exit

In a perfectly competitive market, no single buyer or seller can influence the market price.

5. What is the difference between supply and stock?

- **Supply** refers to the quantity of a good or service that producers are willing and able to offer for sale at a given price. It is a function of various factors, including production costs, technology, and market conditions.
- **Stock** refers to the total quantity of a good or service that is available in the

market at a given time. It includes both the supply and any inventories held by producers or consumers.

6. Differentiate between Total & Marginal Utility.

- **Total Utility** is the total satisfaction a consumer derives from consuming a certain quantity of a good or service.
- **Marginal Utility** is the additional satisfaction a consumer derives from consuming one more unit of a good or service.

7. How can inflation be controlled in Pakistan?

Pakistan can control inflation through a combination of **monetary and fiscal policies**.

- **Monetary Policy:** The State Bank of Pakistan (SBP) can raise interest rates to make borrowing more expensive and reduce the money supply, which can help curb inflation.
- **Fiscal Policy:** The government can reduce its spending and increase taxes to decrease aggregate demand and slow down the economy, thereby reducing inflationary pressure.

8. What is the difference between GNP & GDP?

- **Gross National Product (GNP)** measures the total value of goods and services produced by a country's residents, regardless of where they are located. It includes income earned by citizens working abroad.

- **Gross Domestic Product (GDP)** measures the total value of goods and services produced within a country's borders, regardless of who owns the factors of production.

9. What is fiscal policy? Write down its objectives.

Fiscal policy is the use of government spending and taxation to influence the economy.

Its primary objectives include:

- **Economic growth:** Stimulating economic activity and creating jobs/
- **Price stability:** Controlling inflation and preventing deflation.
- **Full employment:** Reducing unemployment rates.
- **Income distribution:** Ensuring a fair distribution of income and wealth.

10. What is the Output method to measure National Income?

The **Output method** of measuring national income calculates the total value of final goods and services produced in an economy during a given period. It involves summing up the value added at each stage of production, from raw materials to the final product.

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B.S. in Computer Science Third Semester / Fall 2022

Introduction to Economics

CourseCode:UE-171

Time: 3 Hrs.

Marks:60

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Short Answer Questions (6 marks each)

11. Explain the concept of trade cycle.

The **trade cycle** (also known as the business cycle) refers to the fluctuations in economic activity over time, characterized by periods of growth, contraction, recession, and recovery. It is a natural phenomenon that occurs in most market economies.

The business cycle consists of six stages that describe the fluctuations in economic activity over time:

1. **Expansion:** Economic growth increases, with rising employment, wages, production, and demand. Investment is high, and debts are paid on time.
2. **Peak:** Growth reaches its maximum. Prices and other economic indicators are at their highest, signaling a potential shift in the economy.
3. **Recession:** Demand starts to fall, leading to a decline in production. Prices drop, and economic indicators like income and output decrease.
4. **Depression:** Unemployment rises, and the economy shrinks further, moving below normal growth levels.
5. **Trough:** The economy hits its lowest point, with minimal demand, supply, and national income.
6. **Recovery:** The economy begins to improve. Demand and production pick up, leading to rising employment and investment until steady growth is restored.

This cycle repeats as the economy moves through periods of growth and decline.

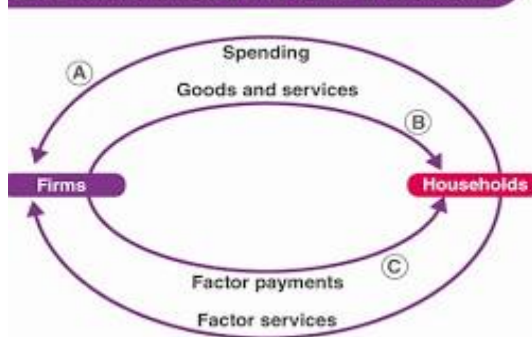
12. Explain any two features of indifference curve.

An **indifference curve** represents all combinations of goods and services that yield the same level of utility to a consumer. Two key features of indifference curves are:

- **Downward slope:** Indifference curves slope downward, reflecting the assumption that consumers prefer more of both goods.
- **Convexity:** Indifference curves are typically convex to the origin, indicating that consumers are willing to trade off some of one good for more of another, but at a decreasing rate.

13. Draw and label the circular flow of income and product.

CIRCULAR FLOW OF INCOME IN A SIMPLE ECONOMY



The **circular flow of income and product** illustrates the interconnectedness of economic activities in a market economy. It shows how households provide factors of production (labor, land, capital, and entrepreneurship) to firms, which use these inputs to produce goods and services. Households receive income (wages, rent, interest, and profits) from firms, which they use to purchase goods and services. This

spending creates demand for goods and services, leading firms to produce more.

14. Define economic growth.

Economic growth refers to an uninterrupted increase in the production of goods and services in an economy over time. It is typically measured by the percentage change in real GDP. Economic growth is essential for improving living standards, reducing poverty, and creating jobs.

15. Explain cross elasticity of demand.

Cross elasticity of demand measures the responsiveness of the demand for one good to a change in the price of another good. It is calculated as the percentage change in the quantity demanded of good A divided by the percentage change in the price of good B.

$$(\%d(Q_a)) / (\%d(P_b))$$

- If the cross elasticity of demand is positive, the two goods are substitutes (e.g., coffee and tea). When the price of one good increases, the demand for the other good increases.
- If the cross elasticity of demand is negative, the two goods are complements (e.g., cars and gasoline). When the price of one good increases, the demand for the other good decreases.

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Instructions: The answers must be attempted on the answer sheet provided.

Short Answer Questions (3 marks each)

16. Labor Participation Rate

The **labor participation rate** is the percentage of the working-age population that is either employed or actively seeking employment. It measures the portion of the population that is available for work.

17. Indifference Curve

An **indifference curve** represents all combinations of goods and services that yield the same level of utility to a consumer. It is used in consumer theory to analyze consumer preferences and choices.

18. Factors of Production

Factors of production are the resources used in the production process. They are typically classified into four categories:

- **Land:** Natural resources used in production, such as land, minerals, and water.
- **Labor:** Human effort and skills used in production.
- **Capital:** Man-made goods used in production, such as machinery, equipment, and buildings.
- **Entrepreneurship:** The ability to combine the other factors of production to

create new goods and services.

19. Fixed Cost

Fixed costs are costs that do not vary with changes in output. They remain constant regardless of the level of production. Examples of fixed costs include rent, salaries of top management, and property taxes.

20. Marginal Revenue

Marginal revenue is the additional revenue a firm earns from selling one more unit of output. It is calculated as the change in total revenue divided by the change in quantity sold.

21. Elasticity

Elasticity measures the responsiveness of one variable to changes in another variable. In economics, it is often used to measure the responsiveness of quantity demanded or supplied to changes in price or income.

- **Price elasticity of demand** measures how responsive quantity demanded is to changes in price.
- **Income elasticity of demand** measures how responsive quantity demanded is to changes in income.

22. Total Utility

Total utility is the total satisfaction a consumer derives from consuming a certain quantity of a good or service.

23. Monetary Policy

Monetary policy refers to the actions taken by a central bank to influence the money supply and interest rates in an economy. It is used to achieve macroeconomic objectives such as price stability, full employment, and economic growth.

24. Real Gross Domestic Product (Real GDP)

Real GDP is a measure of economic output that is adjusted for inflation. It provides a more accurate picture of economic growth by accounting for changes in prices.

25. Purpose of Economics

The **purpose of economics** is to understand and explain how individuals, businesses, and governments make choices about allocating scarce resources. It aims to provide insights into economic behavior, analyze economic problems, and develop policies to improve economic outcomes.

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B.S. 4 Years Program / Eighth Semester-Spring 2022

Paper: Introduction to Economics

Time: 3 Hrs

Course Code: IT-408

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED
Q.1. Write short answers to the following questions.

1. Define the following (10 x 2 = 20)

Short Answer Questions

26. Price Floor

A price floor is a minimum price set by the government for a particular good or service. It is typically set above the equilibrium price, leading to a surplus of the good.

27. Shift in Demand

A shift in demand occurs when consumers change their willingness or ability to buy a good or service at a given price. This can be caused by factors such as changes in income, preferences, population, or the price of related goods.

28. Monopoly

A monopoly is a market structure where there is only one seller of a good or service. The monopolist has significant market power and can set prices above the competitive level.

29. Elasticity

Elasticity measures the responsiveness of one variable to changes in another variable. In economics, it is often used to measure the responsiveness of quantity demanded or supplied to changes in price or income.

- Price elasticity of demand measures how responsive quantity demanded is to changes in price.
- Income elasticity of demand measures how responsive quantity demanded is to changes in income.

30. Microeconomics

Microeconomics focuses on the behavior of individual economic agents, such as consumers, producers, and firms. It examines how markets work, how prices are determined, and how resources are allocated.

31. Consumer Price Index (CPI)

The Consumer Price Index (CPI) measures the average change in prices paid by consumers for a basket of goods and services. It is used to track inflation and adjust wages and pensions for rising living costs.

32. Fiscal Policy

Fiscal policy is the use of government spending and taxation to influence the economy. It is a tool used by governments to achieve macroeconomic objectives such as economic growth, price stability, full employment, and income distribution.

33. Unemployment Rate

The unemployment rate is the percentage of the labor force that is actively seeking employment but unable to find a job. It is a key indicator of the health of an economy.

34. Factors of Production

Factors of production are the resources used in the production process. They are typically classified into four categories:

- Land: Natural resources used in production, such as land, minerals, and water.
- Labor: Human effort and skills used in production.
- Capital: Man-made goods used in production, such as machinery, equipment, and buildings.
- Entrepreneurship: The ability to combine the other factors of production to create new goods and services.

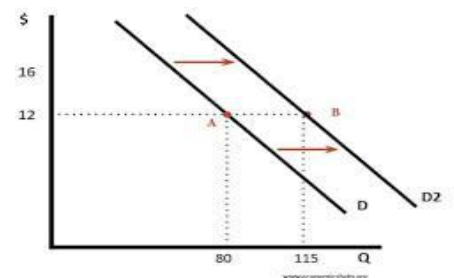
35. GDP per Capita

GDP per capita is a measure of economic output per person in a country. It is calculated by dividing the country's GDP by its population. GDP per capita is often used to compare living standards across different countries.

2. Draw diagrams for the following. (5 x 2 = 10)

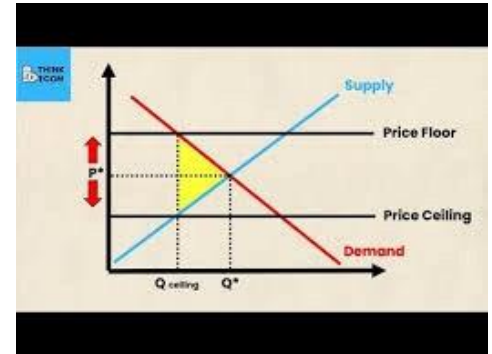
1. Positive Shift in Demand

When demand increases, the demand curve shifts to the right. This leads to a higher equilibrium price and quantity.

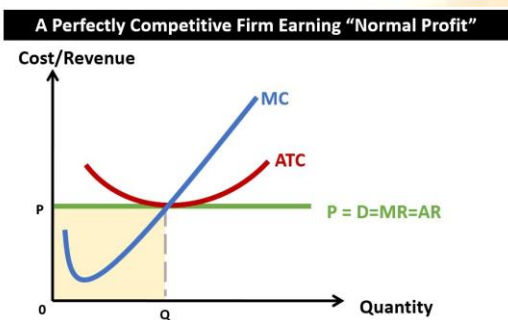


2. Price Ceiling

A price ceiling is a maximum price set by the government below the equilibrium price. It leads to a shortage of goods as the quantity demanded exceeds the quantity supplied.



3. Normal Profit for Perfect Competition



In a perfectly competitive market, a firm earns a normal profit in the long run. This means that its total revenue is equal to its total cost, and the firm is just breaking even.

MR (Marginal Revenue) = Price (P)

AR (Average Revenue) = P

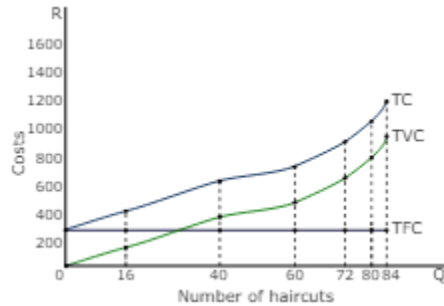
ATC (Average Total Cost) = (Total Fixed Cost + Total Variable Cost) / Quantity

MC (Marginal Cost) = $\Delta \text{Total Cost} / \Delta \text{Quantity}$

D (Demand) = P = MR

$P = MR = AR = MC = ATC$

4. Fixed Cost, Variable Cost, and Total Cost in the same graph

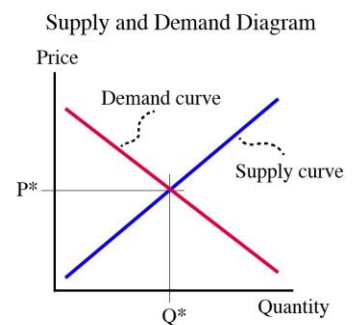


- Fixed cost (FC): A horizontal line, as it remains constant regardless of output.
- Variable cost (VC): An upward-sloping line, as it increases with output.
- Total cost (TC): The sum of fixed and variable

costs. It is also upward-sloping but starts higher than VC due to the fixed cost component.

5. Market Equilibrium

Market equilibrium occurs where the demand curve intersects the supply curve. At this point, the quantity demanded equals the quantity supplied, and there is no tendency for the price to change.



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Subject: Introduction to Economics

Paper: UE-171 Time: 3 Hrs

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

Short Answer Questions

1. Differentiate microeconomics from macroeconomics.

- **Microeconomics** focuses on the behavior of individual economic agents, such as consumers, producers, and firms. It examines how markets work, how prices are determined, and how resources are allocated.
- **Macroeconomics** deals with the economy as a whole, analyzing aggregate variables like GDP, inflation, unemployment, and interest rates. It studies the factors that influence the overall economic performance of a nation.

2. Explain the cost of production as a factor of supply.

The **cost of production** is a key determinant of supply. Producers are generally willing to supply more of a good or service at higher prices to cover their increasing costs. As production costs rise due to factors like higher input prices, technological changes, or changes in regulations, producers may supply less of the goods at a given price, causing the supply curve to shift leftward.

3. Define the Production Possibility Frontier with an example.

The **Production Possibility Frontier (PPF)** is a graphical representation of the maximum combination of two goods or services an economy can produce with its existing resources and technology, assuming those resources are fully and efficiently utilized.

Example: A country with limited resources can either produce agricultural goods or manufactured goods. The PPF shows the maximum possible combinations of these two goods that the country can produce. If the country chooses to produce more agricultural goods, it will have to produce fewer manufactured goods, and vice versa.

4. Discuss Government Expenditures as a component of GDP.

Government expenditures are a significant component of GDP, representing the total spending by governments on goods and services. This includes spending on infrastructure, education, healthcare, defense, and other public services. Government expenditures contribute to economic activity by creating jobs and stimulating demand for goods and services. However, excessive government spending can lead to budget deficits and inflation.

5. Explain the relationship between interest rate and savings.

The **interest rate** and **savings** have a generally inverse relationship. When interest rates are higher, people tend to save more because they can earn a higher return on their savings. Conversely, when interest rates are lower, people may be less incentivized to save and instead opt to spend or invest their money. This relationship is influenced by factors like inflation expectations and consumer confidence.

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Paper: Introduction to Economics Course Code: UE-171 Time: 3 Hrs Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (6 x 5 = 30)

1. Differentiate between Monopoly and Monopolistic Competition.

Monopoly:

- Single seller
- No close substitutes for the product
- Significant market power
- Barriers to entry

Monopolistic Competition:

- Many sellers
- Differentiated products
- Limited market power
- Relatively easy entry and exit

2. Explain Unitary Elastic Demand and Perfectly Elastic Demand with the help of an example.

Unitary Elastic Demand:

- When a change in price leads to an equal percentage change in quantity demanded.
- Example: If the price of a product increases by 10%, and the quantity demanded decreases by 10%, the demand is unitary elastic.

Perfectly Elastic Demand:

- When a small change in price leads to an infinite change in quantity demanded.

- Example: In a perfectly competitive market, if a firm raises its price above the market price, consumers will simply buy from other firms at the lower price.

3. Contrast between the Expansion and Contraction stages of an economy.

Expansion:

- Economic activity is growing
- GDP is increasing
- Unemployment is falling
- Inflation may rise

Contraction:

- Economic activity is slowing down
- GDP is decreasing
- Unemployment is rising
- Inflation may fall

4. What will happen to Market Equilibrium in the case of Price Ceiling?

A **price ceiling** is a maximum price set by the government below the equilibrium price.

When a price ceiling is imposed, it creates a shortage as the quantity demanded exceeds the quantity supplied. This can lead to:

- **Excess demand:** Consumers may not be able to purchase the desired quantity of the good.
- **Black markets:** May emerge as sellers try to sell the goods at higher prices than

the ceiling.

- **Reduced quality:** Producers may reduce the quality of the good to cut costs.

5. Write a brief note on Technology as a factor of Supply.

Technology is a significant factor of supply. Advancements in technology can increase productivity and reduce production costs, leading to a rightward shift in the supply curve. This can result in lower prices and increased output.

6. List the Functions of Money.

- **Medium of exchange:** Money is used to facilitate transactions and exchange goods and services.
- **Unit of account:** Money serves as a common unit of measurement for valuing goods and services.
- **Store of value:** Money can be held and used to purchase goods and services in the future.
- **Standard of deferred payment:** Money is used to make promises to pay in the future, such as loans and contracts.
- **Liquidity:** Money is easily convertible into other assets.

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Q.1. Answer the following short questions. (5 x 6 = 30)

1. Explain the concept of utility.

Utility is a measure of the satisfaction or happiness that a consumer derives from consuming a good or service. It is a subjective concept that varies from person to person and can be influenced by factors such as preferences, income, and prices.

2. What is meant by allocative efficiency?

Allocative efficiency occurs when resources are distributed in a way that maximizes the overall satisfaction or welfare of society. This happens when the goods and services produced reflect consumer preferences, meaning the quantity produced is exactly what society desires.

In a perfectly competitive market, allocative efficiency is achieved when:

- **Price (P) = Marginal Cost (MC) (neither profit nor loss)**

At this point, the price consumers are willing to pay for a good matches the cost of making one more unit of it. This ensures that resources are allocated in the most efficient way to maximize societal satisfaction, without any underproduction or overproduction.

3. Discuss the input market in the circular flow of income and product.

The **input market** is the market where firms purchase the factors of production (labor, land, capital, and entrepreneurship) that they need to produce goods and services. In the circular flow of income and product, households supply these factors of production to firms, and firms pay households for their services. The input market plays a crucial role in determining the costs of production for firms and the income earned by households.

4. Explain the effects of a negative shift in demand on the market price of a product.

A **negative shift in demand** occurs when consumers become less willing or able to buy a good or service at a given price. This can be caused by factors such as a decrease in income, a change in preferences, or an increase in the price of complementary goods. When demand decreases, the demand curve shifts to the left. This leads to a **lower equilibrium price** and a **lower equilibrium quantity** of the product.

5. Differentiate between total utility and marginal utility.

- **Total utility** is the total satisfaction a consumer derives from consuming a certain quantity of a good or service.
- **Marginal utility** is the additional satisfaction a consumer derives from consuming one more unit of a good or service.

As a consumer consumes more units of a good, the marginal utility they derive from each additional unit typically decreases due to the **law of diminishing marginal utility**. This means that the satisfaction gained from each additional unit diminishes over time.

Best of luck

